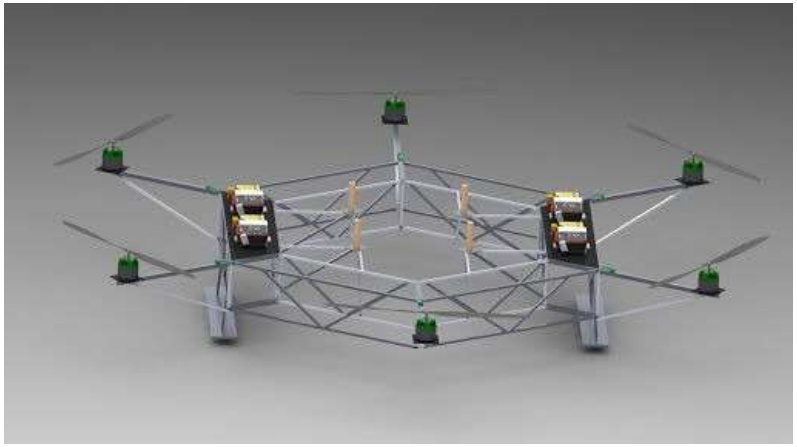




Learning GNC through Animation!!

BY: AKRITI

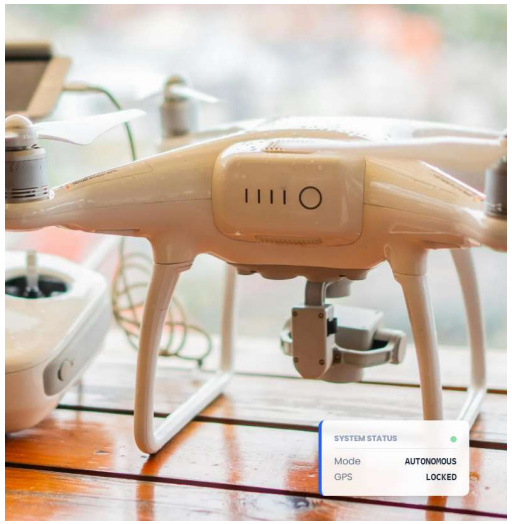
INDIA SPACE LAB

Rules of the Session!!

- This session will focus on the practical aspect than theory
- **If you have Laptop**: You need to open Google Colab from your website; you require only Gmail account; no download required; link of the code will be sent on chat window as per requirement
- **If you don't have Laptop**: take a notebook and pen and understand the process of GNC

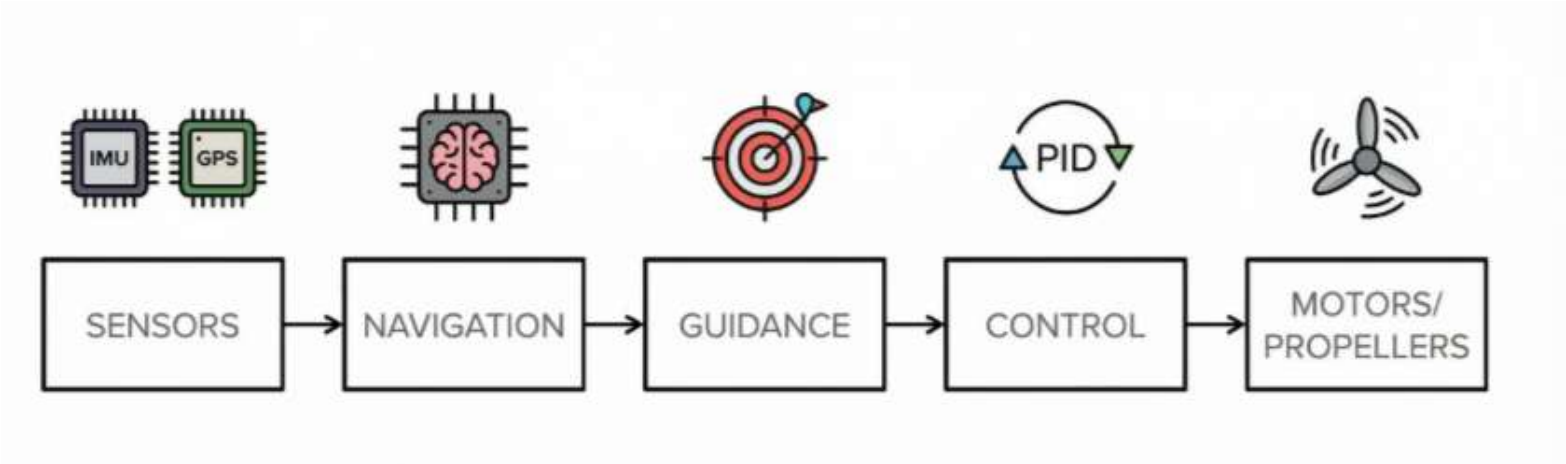
Happy Learning!! ☺

Section 1



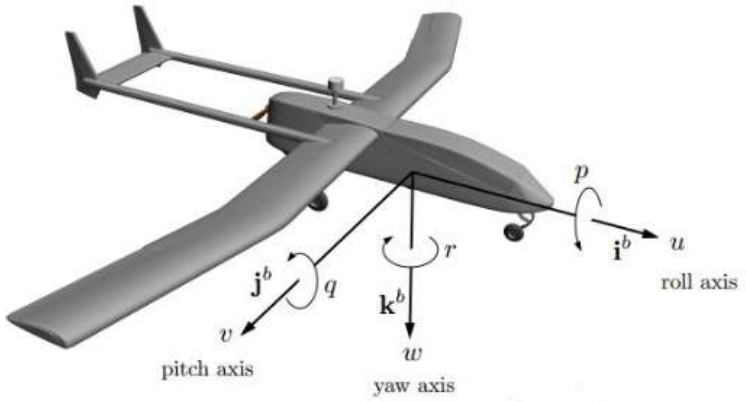
Revisiting the World of GNC

GNC Story



Navigation tells you where you are, **Guidance** tells you where to go, **Control** makes you move…

Navigation



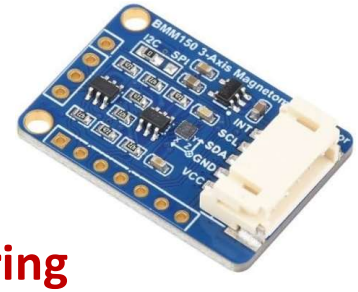
Knowing the **"states"** of the drone

Position	Velocity
Orientation	Angular Rate

IMU



Magnetometer

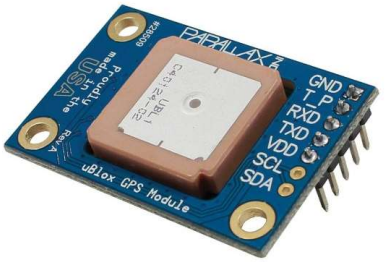


Measuring "States"

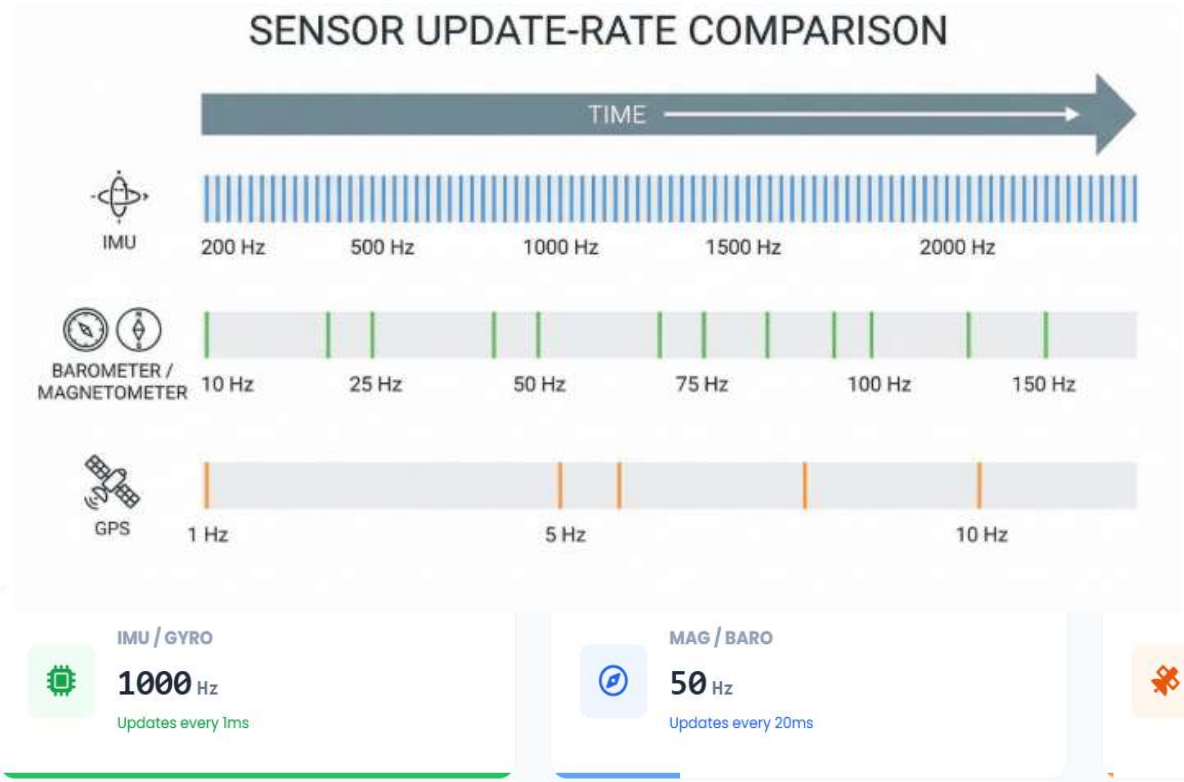
Barometer

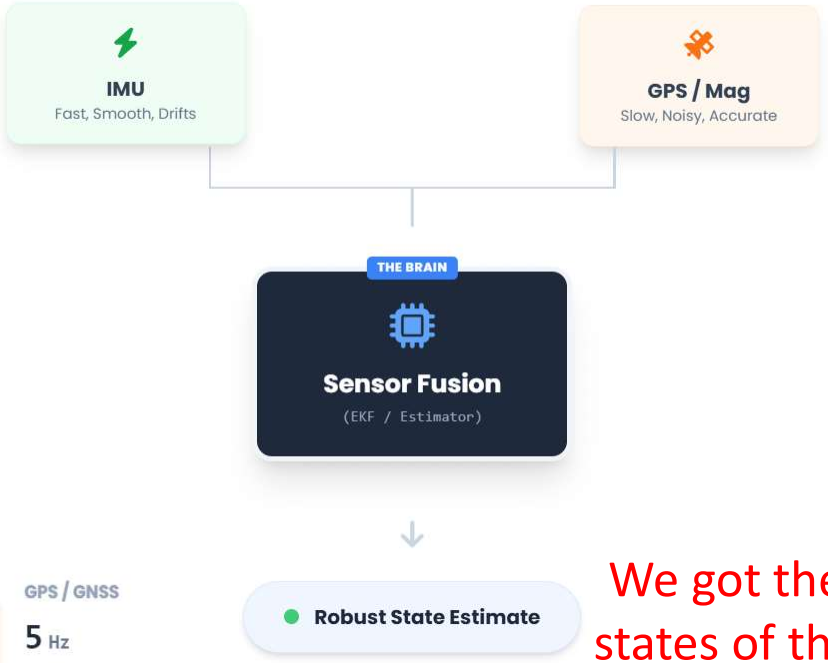


GPS



Navigation

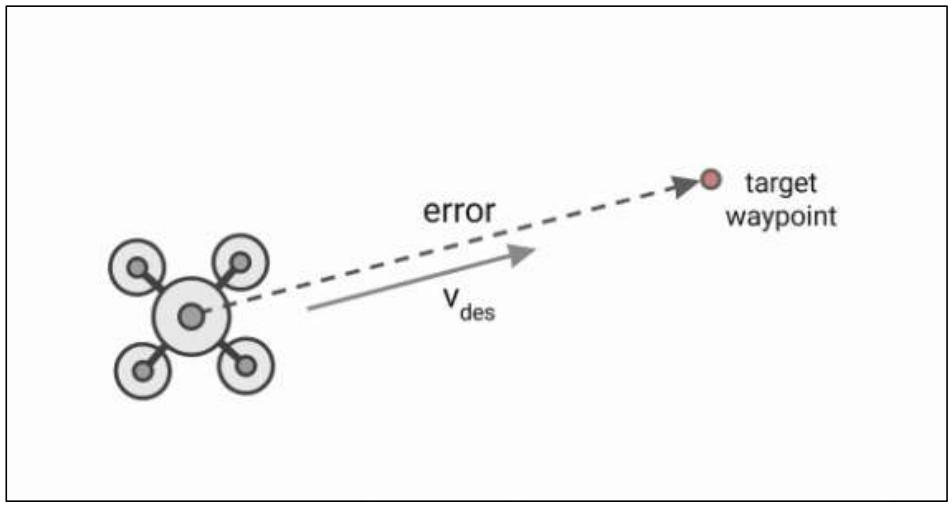






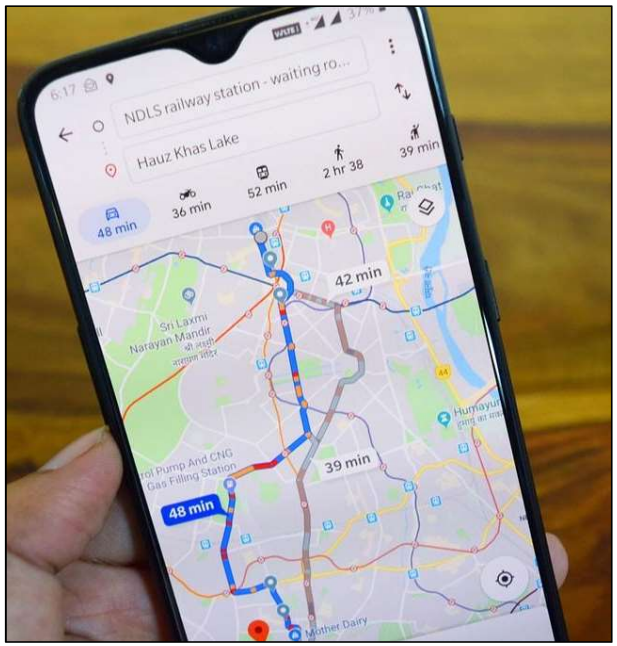
We got the states of the drone!!

Guidance



The core logic is simple: **Error** = **Target** − **Current**.

The desired velocity vector points directly along this error arrow.



"Guidance is the strategy builder”

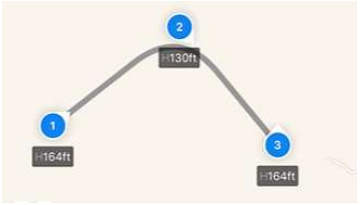
Guidance



Hover at Point

(Position Hold)

Maintains fixed XYZ coordinates. Guidance outputs zero velocity reference.



Fly to Waypoint

(Target Seeking)

Generates velocity vector pointing from current position to target (A → B).



Follow Line

(Path Following)

Minimizes cross-track error to stay on a straight path between two points.



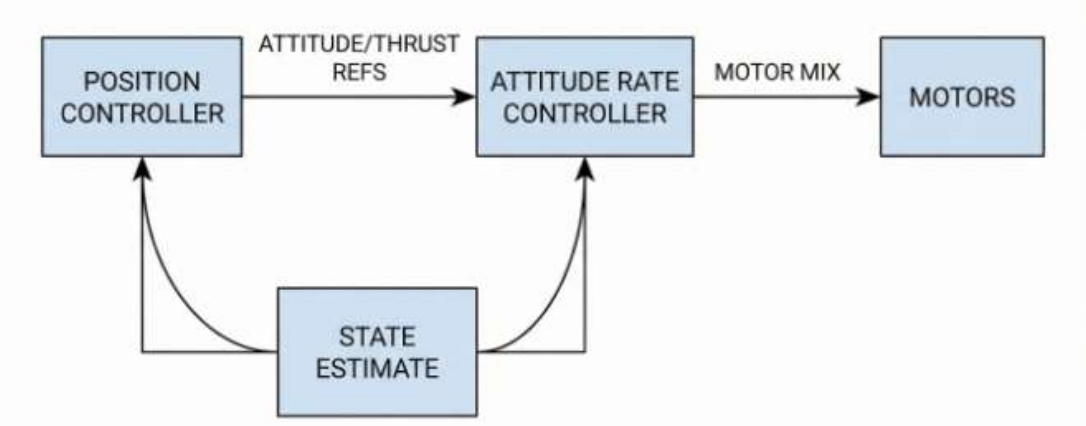
Maintain Alt

(Z-axis Hold)

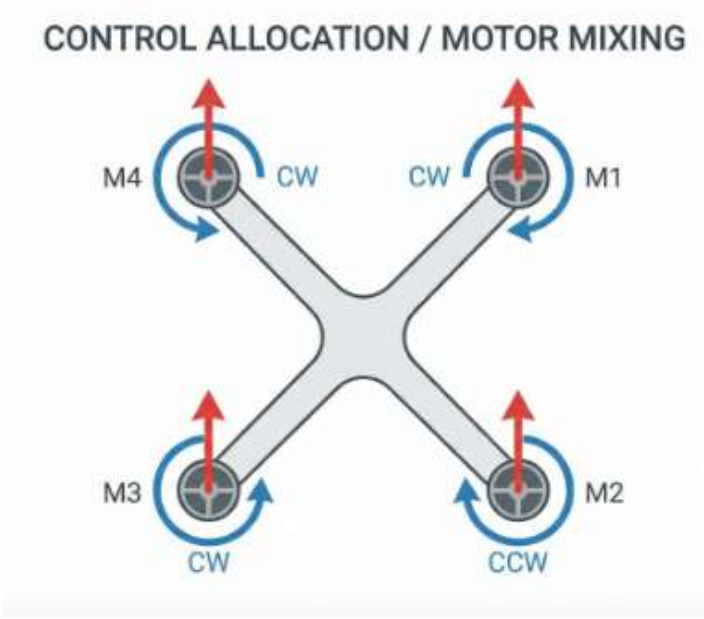
Adjusts vertical velocity to keep barometer/GPS height constant.

We got the desired states of the drone!!

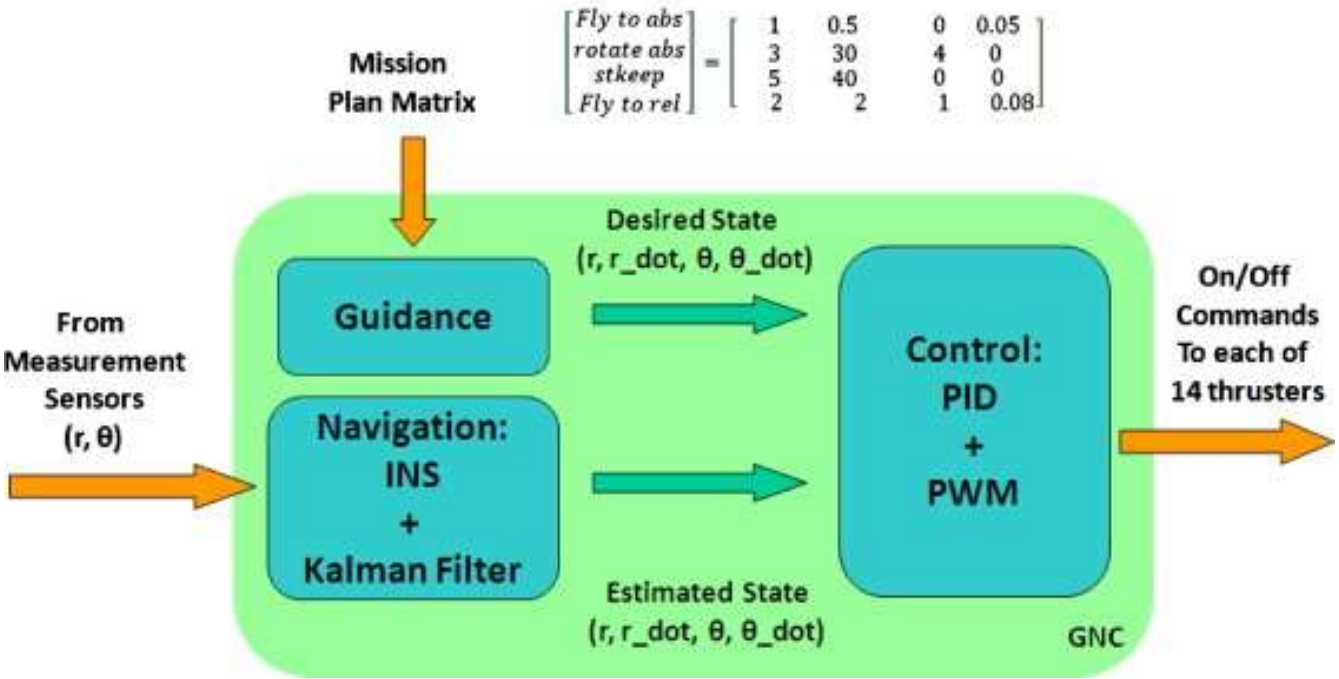
Control



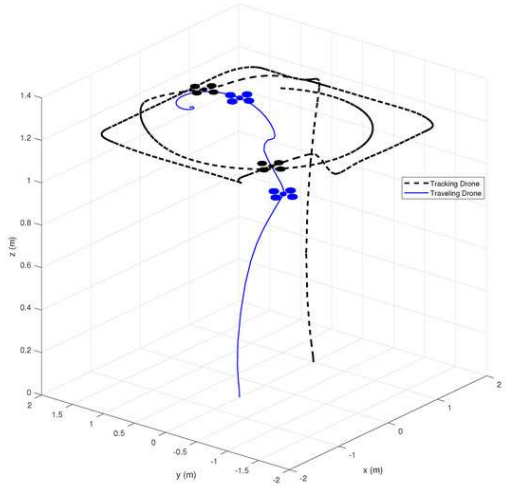
Outer **Position/Velocity** loops cascade references down to inner **Attitude/Rate** loops



GNC Integration

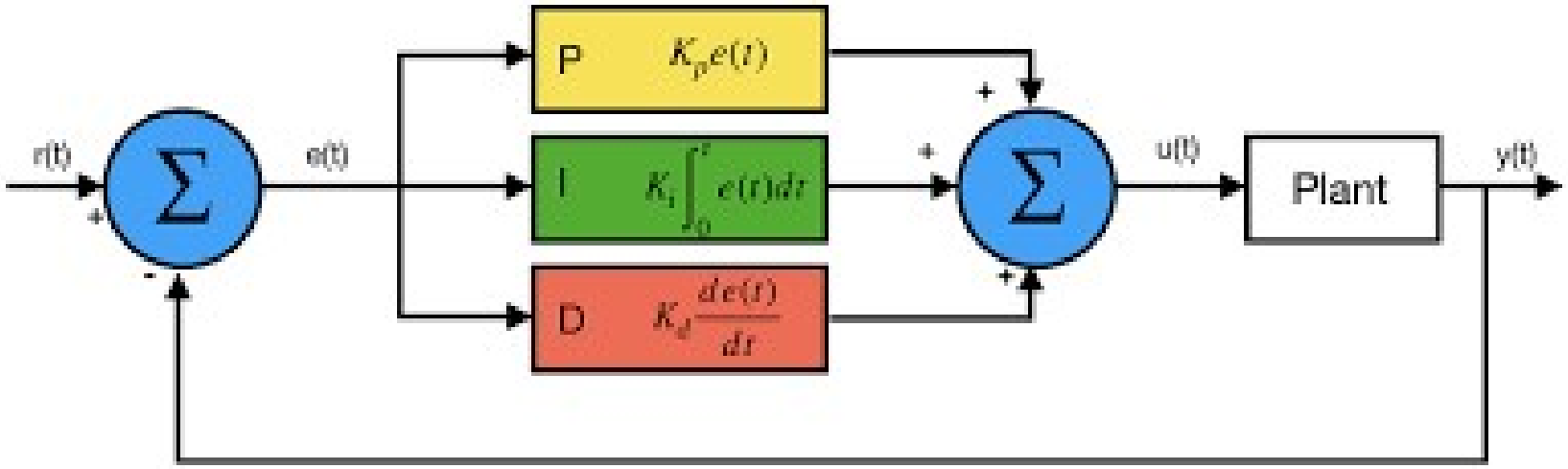


Section 2

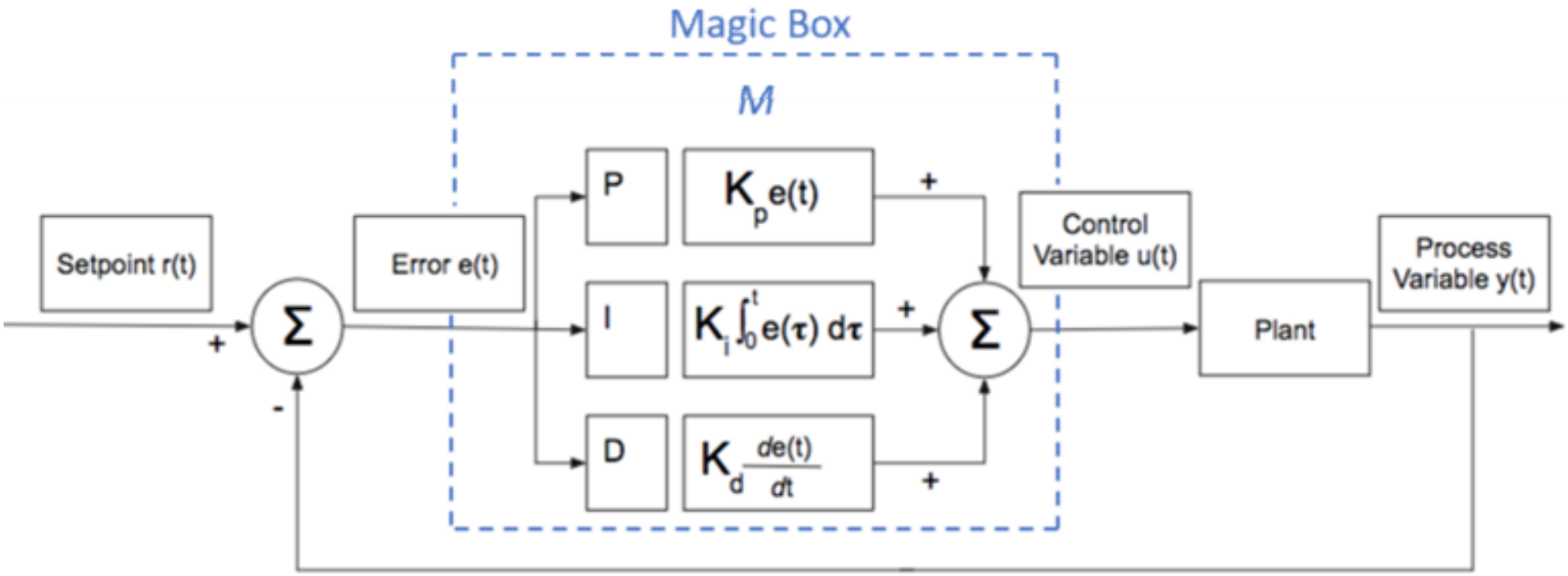


Playing with GNC

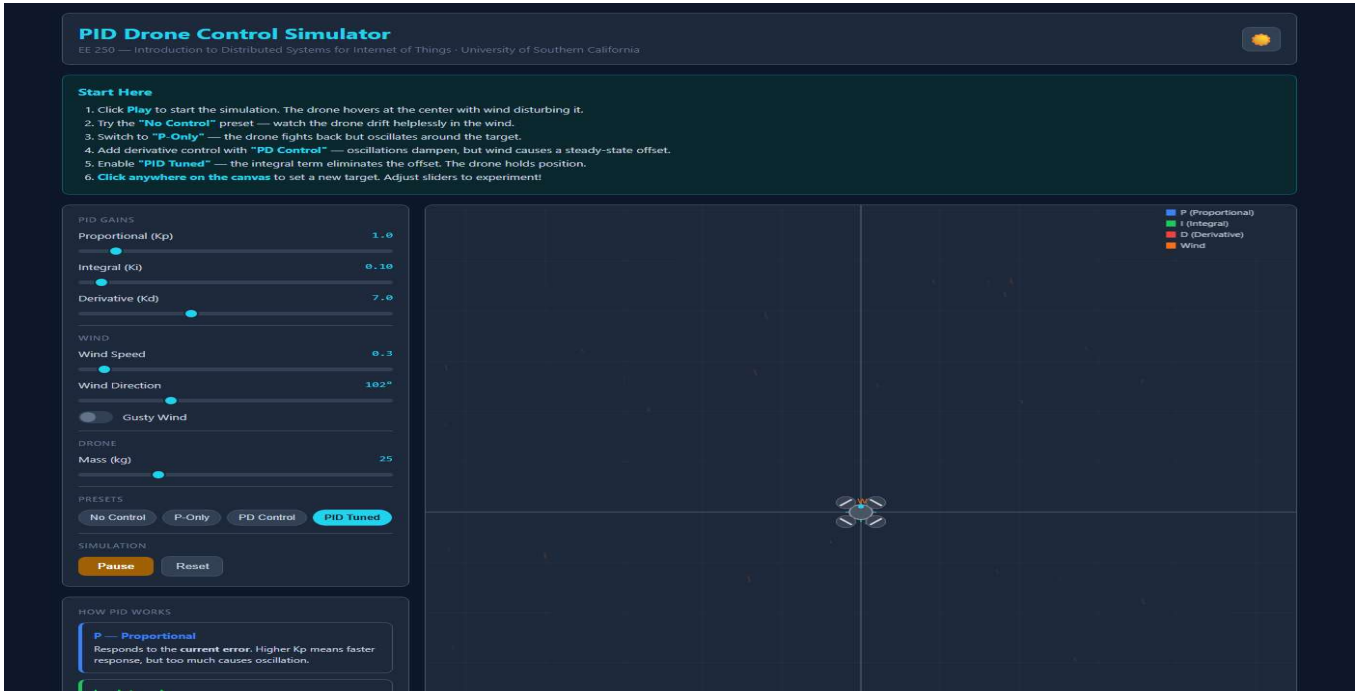
PID Controller



PID Controller



PID Simulator



https://pid-drone-control.vercel.app/

Thank You ☺